

Simulating Public Online Discourse using Large Language Models

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Abstract

The present study aims to further our understanding of Large Language Models’ (LLMs) capabilities of simulating human participants as subjects of social science research. Building on previous work, this paper introduces a customized framework for the simulation of threads – topic discussions – on the social media platform Reddit, using LLMs. The simulation framework unifies statistical models of Reddit user behavior and the generative capabilities of LLMs to govern the decision-making and behavior of simulated agents.

Using demographics and survey data, ranging from self-reported online behavior to knowledge of and interest in diverse political topics, this study created a sample of 180 unique simulation agents. These agents were then divided into two subgroups and set to discuss 20 different political topics. In order to judge the framework’s ability to faithfully reproduce human behavior, given the context, its performance on three key aspects was analyzed: the ability to simulate different individual users, the simulated patterns of interaction, and the recreation of Reddit-specific discussion network structures. According to these three aspects, the simulated discussions were analyzed and compared to empirical data generated by a recent experiment on Reddit. The simulated agents were shown to over-portray human activity with regard to commenting behavior and to struggle with accurately memorizing the discussion history. At the same time, the simulation was able to reproduce temporal patterns of human social media interaction. With the present work, I explore the current possibilities of using LLMs as subjects in the simulation of human online discourse.

1 Introduction

Is it possible to create a simulation of social media discourse that realistically mimics human behavior and produces tangible output in the form of believable discussions? Social media plays an integral part in the lives of many, with platforms like Facebook, X, and Reddit counting billions of users worldwide. While social media offers to enrich its users’ lives by enabling them to connect with like-minded others, keep in touch with each other, and so on, it has also gained a certain notoriety in recent years. The prevalence of anti-social behavior like discrimination, sexism, and toxicity on these platforms is well established [36, 42, 44]. As are the formation of echo chambers where users are shielded from opposing views [8, 12] and the prevalence of fake news [24]. This, in combination

with social media being increasingly employed for political purposes [47], makes studying human behavior on these platforms and gaining an understanding of the mechanisms guiding it ever more important. With platforms increasingly limiting and regulating access to their data and the ethical as well as practical constraints involved in conducting social media experiments, especially those investigating anti-social behaviors, the simulation of social media, if successful, promises an alternative. As user-generated text content is a major part of social media, any simulation hoping to claim validity must therefore also simulate it. For this purpose, Large Language Models (LLMs), trained for the generation of human-like text, seem to offer themselves as the perfect simulation agents. LLMs have become so good at a myriad

of tasks that it has become hard to tell the difference between artificial and human content [11]. Research has also shown that LLMs are capable of replicating empirical survey results when asked to answer political [4] and psychological [38] surveys. In the present work, I therefore propose a simulation framework aimed at harnessing LLMs’ capabilities to reproduce human behavior. The presented framework is designed to facilitate the simulation of discussions on the social media platform Reddit, using LLM-based agents that dynamically choose what actions they take and which content they generate. To my knowledge, this constitutes the first advance of its kind in this specific direction.

I test the framework’s performance by simulating discussions of 20 different political topics among 180 agents. The respective simulation design was inspired by a recent experiment on Reddit [34] and closely follows its design. The simulated discussions are analyzed, and the results are compared to those derived from the experimental data. In order to assess the capability of the simulation framework to mimic human social media behavior, the results are analyzed according to three key aspects:

1. **Persona imprinting:** What aspects of human behavior do the simulated agents reproduce, and if so, how closely do they capture the behavior?
2. **Social media interaction:** Do the agents believably mimic interaction patterns typically observed within social media and especially Reddit?
3. **Discussion networks:** Are agents able to recreate the structure of Reddit discussion networks observed in empirical data?

The framework has been made publicly available online with the goal of providing other researchers with the necessary tools needed to run their own simulations. So has all the data needed to reproduce the findings of this work. See 8 for the supplementary information (SI).

2 Related Research

2.1 LLM Research

The potential benefit of integrating LLMs into social scientific research has caused the emergence of many forays and investigations into this area.

2.1.1 Generative Agent Based Models

Generative agent-based models (GABMs) promise to unify the advantages of classical agent-based models (ABMs) and LLMs. The study of social (media) networks, especially, seems determined to harness the possible potential of GABMs. Research into this area is manifold [13, 14, 19, 48].

GABMs have already been used to study the influence of different kinds of feed algorithms on social media discourse within a simulation of social media platforms [48]. Other work has employed the concept to study the emergence of echo chambers, validating their results by comparing their GABMs to a real social media data [13, 19]. Phenomena like emotion, opinion, and information propagation have also been studied by incorporating LLMs into an agent-based simulation framework and using real-world social network data for validation [14]. Others have implemented LLMs into complex simulations where the models not only performed a single task – like text generation or classification – but were tasked to perceive environments, choose how to act in given situations, and finally perform the self-determined action [26, 37, 54]. While this rich background of research utilizing GABM frameworks evokes a certain optimism regarding GABMs’ possible applications, other work showed that researchers must be aware of the perils that come with the inclusion of LLMs into the classical ABM concept [23]. Generalization and especially validation remain key aspects that must not be forgotten when considering employing GABMs to simulate human behavior [23].

2.1.2 Persona Imprinting

The concept of LLM persona imprinting is used to elicit a broad range of different behaviors from LLMs. On top of the task to be completed, LLMs are therefore given specific, real or fictive, persona data upon which to base their responses. This persona data can take the form of a simple point of

view [21], demographic data [39], survey data [4], a short persona description [35, 37], or interview data [38].

Persona imprinting using socio-demographic persona data has been used to show that LLMs replicated the voting behavior of their sub-population, possibly even over different moments in time [4]. Other work has used extensive qualitative interviews and revealed that LLMs were able to reproduce individual-level social survey responses given by the interviewed human participants [38]. It has also been uncovered that LLMs display emotional reactions governed by gender stereotypes when prompted with differently gendered persona data [39].

While the concept of imprinting LLMs with specific personas is tempting, one must also be aware of possible concerns. Sociopolitical persona-based LLM-generated survey responses showed significantly less variation and different regression coefficients than the human baseline data [9]. On top of that, LLMs were shown to misrepresent demographic sub-populations through out-group misportrayal, group flattening, and treating demographic information as unchanging and innate characteristics of individuals [49]. Considering these risks, one must critically reflect on the outputs produced by LLMs in general. Keeping this in mind, it is important to restate that the present work is not trying to replace humans as research subjects but to augment social media research where human involvement might be costly, unfeasible, or outright harmful.

2.2 Social Media Research

In order to robustly validate the presented framework, it is crucial to understand key elements of human online behavior. The present work focuses on interaction patterns and emergent discussion structures.

2.2.1 Social Media Interaction

Human online communication takes place in burst-like patterns, with shorter windows of increased activity separated by longer periods of silence [10, 30, 50–52]. These burst-like patterns occur when the time between subsequent interactions, the so-called inter-event time (IET), follows a heavy-tailed distribution. Independent Poisson

processes have been shown to model the first interaction – i.e., the initiation of the individual bursts – well [52]. This means that the bursts of intense interaction occur at a constant average rate. The bursts themselves, on the other hand, were discovered to be non-Poissonian [51]. The individual actions within a burst, therefore, do not take place at a constant rate and are independent from each other. IET of human interaction has been shown to follow a power-law distribution in the study of short messages [50, 52] and commenting behavior on social media platforms [30, 51, 56]. Other research has shown the distribution to follow an exponential shape [27] on a social media platform, where previous research found it to be power-law shaped [52]. Yet others argue for the IET distribution of online interaction to be log-normal shaped, especially for actions where instant reaction isn’t possible [10]. Commenting is one possible action where this applies, as it typically requires more time to create a comment than to give a like, for example. Possible factors responsible for the non-Poissonian distribution of IET within action bursts have been discovered to include individual decision-making processes [52], sometimes linked to topic attraction [56] and user preference selection [30].

Another important aspect of social media interaction is participation. Which users decide to contribute, and who remains silent? The majority of users only view content on social media, while a minority creates it. This is known as the so-called 90-9-1 rule [5]. Thereafter, 1% of users create new content, 9% actively interact with it by commenting, for example, and 90% of users only view the content. Research has found this general trend to hold true on social media [6, 34, 56]. The reasons why users remain silent have been grouped into environmental, personal, relationship, and security reasons by existing literature [46]. In the context of political online discussions specifically, prior research suggests a two-phase process governing user decision to become active: influenced by their political interest and knowledge, users decide to view political content [33], of those who do, a small group then becomes active [34]. This active minority has been shown to display more extreme opinions than average users [6]. In the context of Reddit, this active minority has been shown to consist of politically interested

males who perceived their environment to be toxic [34].

2.2.2 Discussion Networks

Discussion network structure is typically visualized using graphs. Each node within the graph represents a message, and the edges represent the messages' relations. In the case of thread-based platforms like Reddit, this representation is tree-shaped. There is one root – i.e., an initial post – and each comment is represented by a node. The resulting structure can typically be described by identifying four key characteristics [3]. First, the size of the network, that is, the number of individual messages. The width denotes the maximum number of nodes on any given level of the tree. The distance between the root and its furthest node is known as depth. Finally, scale describes the number of unique message authors. Research has used these key features to validate theoretical models of discussion networks [18, 20]. Others studied the impact of individual- and group-level features, like toxicity, on the emergent structure of the network, using the above aspects to define network structure [44, 55].

3 Background

3.1 tl;dr Reddit

Reddit is a social media platform with many sub-communities, so-called Subreddits or Subs. Each of these Subs is typically dedicated to a topic and has its own rules and guidelines. Subreddits for the same topic, though, are not uncommon.

Reddit is a thread-based platform. This means that almost all user interaction happens in threads. A thread is made up of a single post and any number of comments. It can therefore be described as the discussion that evolves from a post. Posts, by design, must be posted in a specific Subreddit and typically set the topic of the discussion, which lies within the broader topic defined by the Subreddit.

In a literature Sub, posts might be about different books, story characters, or writers, but not about sports teams, hiking equipment, or kittens. Once a post starts to collect comments, which may only be created as replies to the post or a single comment, the respective discussion network

starts to increase its width and/or depth. The tree begins to grow, so to speak. Depending on their location within the depth of the tree, comments may be categorized as first-, second-, third-,... level comments. Figure 1 visualizes this.

Users visiting the platform are presented with their personal feed, a selection of existing threads from different Subreddits. Each thread is represented by the title of the post, and the selection is based on Reddit's recommendation algorithm. From here, users may visit a specific Sub and create a new thread by posting or view an existing one. Alternatively, they may choose to view a thread directly from their feed. Once the users enter the thread view, they can choose to remain silent or become active by liking and disliking content or creating a new comment.

3.2 Baseline Experiment

For the purpose of validating the presented simulation framework, I decided to use the participant and experimental data collected by Oswald et al. [34]. In the following, this will be referred to as the baseline experiment and the baseline data. Oswald et al. conducted a Reddit experiment with 520 participants from the US to investigate participation inequality on the platform. They had participants discuss 20 different political topics in private Subreddits. The participants were split into three experimental conditions: control, incentives, and moderation. Participants in the incentives group were promised monetary recompense for every day on which they would write at least one meaningful comment. Those in the moderation condition were told to be civil and that content would be moderated. Due to missing options for moderating content within the simulation and to avoid overcomplicating the simulation, only the baseline experiment's control condition was simulated. There were 180 participants in the control condition, who were divided into two individual Subreddits.

Over a four-week period, Oswald et al. released 20 different posts. One every working day. These posts consisted of statements regarding political topics, and the participants were asked to discuss the statements. One example of a statement would be: "Fur clothing should be banned.". For the other topics, see Table D2. Over the course of the



Fig. 1: Reddit comment levels and thread structure as visualized by Yu et al. [55]. Note that “Karma” here is synonymous to post/comment score which is defined as the sum of all likes minus the sum of all dislikes.

study, participants were presented with six surveys. A pre-survey, four weekly check-up surveys, and a post-survey. The pre-survey queried participants for demographic information, their online behavior, interest in specific political topics, and their respective knowledge of these topics. Specifically, participants were asked how much time they spent online, how often they used social media, how often they commented on social media, where they would position themselves on the political spectrum, whether they agreed with certain political statements, and how much they knew about the topics the statements pertained to. The individual items typically had to be answered on a five-point scale along the lines of: never, once per month, once per week, once per day, multiple times a day. For the complete pre-survey template, see Appendix A. The weekly check-up surveys would query participants for their perception of the discussions, among other things. The post-survey mostly repeated the items from the pre-survey. Since the check-up surveys and post-survey are nearly irrelevant for the simulation, additional information won’t be provided here.

3.3 OASIS

The presented simulation framework builds upon the recently published OASIS – Open Agent Social Interaction Simulations with One Million Agents framework [54]. OASIS provides modular components for the platform, LLM-based agent, and environment simulation. The platform module provides platform-specific actions such as `like_comment` and `create_comment`. For the full list of actions, see Appendix B. The agent module introduces an LLM into the simulation and handles LLM-based decision-making and content-generation. The environment module initializes the simulation, progresses it, and updates the simulated platform’s database.

In the case of a thread-based platform like Reddit, one can imagine a basic simulation as follows: The platform and underlying database structures are created. LLM agents that simulate human users are signed up, and their persona data is stored as they “join” the platform. Since the platform has just been created, no content yet exists. By manually defining an initial post, the

simulation is nudged in the desired direction. The environment steps are executed one after another. Each step captures the current state of the platform and passes this information to the agents. Along with this information, the agents are also imprinted with their personas and provided with detailed information on all available actions. All this information is passed to the agents in text form. The LLM agents now observe the state of the platform, which might look something like this in the early stages of the simulation:

```
{post_id: 1,
 user_id: 0,
 content: "Social Media is a threat to
  democracy.",
 score: 3
 created_at: "2025-01-07 120012"},
 {comment_id: 1,
  parent_comment_id: None,
  user_id: 3,
  content: "Exactly, finally ...",
  score: 1,
  created_at: "2025-01-07 120420"},
 {comment_id: 3,
  parent_comment_id: 1
  user_id: 6,
  content: "You are absolutely
    right! ...",
  score: 0,
  created_at: "2025-01-07
    121030"},
 {comment_id: 2,
  parent_comment_id: None,
  user_id: 1,
  content: "This is rubbish",
  score: -2,
  created_at: "2025-01-07 120854"}
```

After observing the environment, the agents choose which action to execute based on the text-form description of what these actions do. The agents return the name of the action along with the required parameters. This info is parsed, and the actions are executed as specified by the agents by calling the respective platform functions. Finally, the actions are recorded as database entries, and the state of the platform is updated, which concludes the environment step. The simulation either ends now, or the next environment step is executed. Depending on the number of agents in the simulation and the complexity of the environment, each step might take anywhere between seconds to possibly hours in real time.

OASIS was shown to closely replicate empirical findings on network size and width in a simulation of online information spread. Network depth was found to be significantly shallower. In another simulation, empirical findings on group polarization effects were reproduced. Finally, OASIS was shown to mimic human herd effect behavior, promoting popular and disliking unpopular content, though it overestimated the negative effect [54].

Key limitations of OASIS with regard to the present work, such as lacking prompt quality and missing structural depth in simulations, are explained in detail in the Methods section. So are the implementations introduced to address them.

4 Methods

4.1 Persona Data

The persona data for the LLM imprinting was created using the survey data collected in the baseline experiment¹.

Participant survey responses were matched to the baseline experiment’s pre-survey template and used to imprint the simulation agents with their respective surveys. The following shows a clipped example of this (seed user responses are bracketed by <>):

```
"... How much time do you spend online
 (browsing the web, using social
 media, etc.)? <multiple hours daily
 > How often do you use social media
 , including Reddit? <multiple times
 per day> How often do you write
 comments online (on social media,
 in news outlets' comment sections,
 etc.)? <never> How interested in
 politics are you? <Moderately
 interested> How much do you agree
 with the following statements? The
 government should not forgive
 student loan debt. <rather agree>
 Airbnb should be banned in cities.
 <rather disagree> ... How much do
 you know about the following issues
 ? Student Loan debt <A moderate
 amount> Airbnb <A little >..."
```

A full example persona is available in the SI.

¹ Available at: <https://osf.io/m8g4x/>

4.2 Simulation Design

The simulation followed the experimental design presented in the baseline experiment. For each of the original 180 participants in the control condition, called (seed) users in the following, an LLM-based agent was created, which was imprinted with the persona data of its respective seed user. The 180 agents were divided into two Subreddits, resulting in the same split as in the baseline experiment. The 20 political statements from the baseline experiment, hereafter called seed posts, were adopted and used as starting points for the discussion simulations.

OASIS provides the LLM-based agents with text input that contains all information about the platform environment, available actions, and simulation-specific instructions. Qwen3-14B [53] was used to simulate agent behavior. A list of available actions can be found in Appendix B. For wording of the exact simulation instructions and reasoning on model choice, see SI. While the simulation-specific instructions and descriptions of available actions are unchanging and of fixed length, the platform environment grows with simulation progress and quickly reaches immense lengths. LLMs are not able to process arbitrarily long sequences of text while still keeping track of their contents, though. The maximum input length for each model is given by its context length. To accommodate this limitation, the simulation was split into smaller sub-simulations, each simulating a discussion of one of the seed posts. Each discussion was therefore simulated independently of each other with agents only being able to access the thread of the current simulation. This design choice eliminated the chance to simulate competition between different threads within the same simulation, but it was a necessary one. Even the Qwen3-family [53] LLMs with the highest specified context length, at time of writing, would not have been able to simulate a multitude of threads over any reasonable amount of time. Competition, in this context, refers to the dilemma users face when confronted with a multitude of existing threads. With which do they interact? By giving the agents in the simulation no chance to interact with any thread but the one that was being simulated, they were denied this choice. In order to still simulate a degree of competition between interests, a pre-simulation layer

was introduced. Before the individual thread simulations began, each agent was shown the content of the seed post and asked to either join the conversation or continue looking for another thread. For the exact prompt, see SI. Only those agents which chose to join were then included in the specific thread simulation.

To mimic the release of only one seed post per day in the baseline experiment, the time in each thread simulation was modified accordingly. Each thread simulation was set to continue for 24 steps, which roughly translates to 24 hours in the simulation. The first thread simulation was set to start at noon on the 1st of July, the second simulation at noon on the 2nd of July, and so on. To clarify, this does not mean the simulations took place on those exact days, but rather that the simulation’s internal time was modified to make it appear like that to the agents. This way, the simulation of the first thread would roughly end as the simulation of the second thread began, resulting in a continuous time progression from the beginning of the first until the end of the last thread simulation. In the baseline experiment, seed posts were released randomly each day between 7 am and 7 pm. For the simulation, this would have inevitably led to gaps and overlaps in the time progression. If one simulation started at 7 am and the next at 7 pm, there would have been a 12-hour gap for which no data was simulated. Vice versa, if a simulation started at 7 pm and the next at 7 am, there would have been an overlap of 12 hours.

To add robustness, 10 separate simulation runs were conducted. Each of the 20 seed post discussions was simulated 10 times for each of the two Subreddits, resulting in 400 individual thread simulations.

4.3 Customization

The following details custom implementations made to deal with OASIS’ limitations in the context of the presented simulation design. All LLM prompts have been corrected for grammatical, spelling, and syntactical errors. Additionally, the prompts have been adapted to reflect the simulation design proposed earlier and were designed to elicit social media common behavior from the agents. Specifically, the agents were told to act in accordance with their respective persona data. They were instructed to be original

and expand upon already present content rather than repeat what was already present. Each agent was assigned a unique ID and asked to prioritize replying to comments on any of their previous comments over replying to an unrelated comment, as well as to avoid replying to comments created by themselves. In case the agents decided to create a comment, they were instructed to do so with a roughly fixed number of words. These values were drawn from the distribution of comment length in the baseline experiment, which, in line with research [45], was log-normally shaped. For replies, agents were tasked to mimic the length of the reply’s target comment. Since social media discourse tends to contain irony and sarcasm as well as anti-social behavior like toxicity and trolling [36, 44], the agents were also prompted to include such behavior when they deemed it appropriate. Exact prompt wordings can be found in the supplementary materials.

OASIS was developed with the intention of using Myers-Briggs Type Intention (MBTI) scores as part of the information used for persona imprinting. Since these scores were not available for the participants of the baseline experiment, OASIS’s underlying database structure was slightly modified to account for this. Parts of the agent prompts responsible for persona imprinting that referred to MBTI scores were removed.

Inspecting OASIS’s implementation of the function responsible for creating new comments showed that it wasn’t designed to facilitate the creation of comments referring to other comments. This explains OASIS’s inability to reproduce human-like levels of discussion tree depth. Therefore, a new function was created that allowed agents to specify a parent comment – i.e., a comment to which the new comment replies – when choosing to compose a comment.

One of the actions available to agents during the simulation was `do_nothing`. Since the wording of the function seemed unrealistic, it was renamed to `continue_browsing`. Its description was adapted to convey the semantic difference. Instead of receiving the option to just remain silent upon being presented with the platform environment, agents were now able to choose the action to ignore the current thread in favor of possibly more interesting content. In line with this, a mechanism was implemented that would retire

agents from the simulation if they remained inactive for too many simulation steps. This was done to simulate loss of interest. The mechanism only applied to agents that had commented during at least one previous simulation step already, and the default value was set to 12 simulation steps.

The final custom implementation governed which of the agents that joined a thread simulation were actively monitoring the platform during each simulation step: the activation function $F(A)$. Simply stated, it simulated which agents were online and viewing the thread during the time window covered by a single environment step. If the activation function returned no activated agents for an environment step, the simulation was suspended for roughly 60 seconds, causing the simulation time to proceed by about an hour. During this time, no agent actions were executed. As mentioned earlier, research on online interaction dynamics describes human interaction to follow burst-like patterns of activity, followed by longer windows of downtime. Inspired by these findings [10, 30, 50–52], the activation function consisted of two parameters: the base activation probability pB and the burst probability modifier pM . $F(A)$ is given as:

$$F(A) = \begin{cases} 1 & \text{if } pB + pM \geq T \\ 0 & \text{if } pB + pM < T \end{cases} \quad (1)$$

where:

- pB denotes the base activation probability.
- pM denotes the burst probability modifier.
- T denotes the activation threshold.

The base activation probability was meant to simulate an agent-specific, constant probability of that agent being online at the moment. It was agent-specific because it was modeled according to the self-reported online behavior of the agent’s seed user. Depending on this information, the binomial probability function:

$$P(X \geq k) = 1 - \sum_{i=0}^k \binom{n}{i} pB^i (1 - pB)^{n-i} \quad (2)$$

where:

- X denotes the random variable representing the number of successes.

- k denotes the threshold number of successes.
- n denotes the total number of trials.
- i denotes the index of summation.
- pB denotes the base activation probability.
- $\binom{n}{i}$ is the binomial coefficient.

was solved for pB , with n fixed at 24 and k fixed at 1, 2, and 3, in such a way that $P(X \geq k)$ assumed reasonable values. $P(X \geq k)$, thus, describes the probability of an agent going online at least once, twice, or three times over the course of 24 simulation steps. Exact values for $P(X \geq k)$ are shown in Table C1.

By determining the comment inter-event time – i.e., the time between subsequent comments by the same user – a log-normal distribution could be fit to the distribution of inter-event time. Using the complementary cumulative density function CCDF of the fit, pM modeled an increased but with elapsing time decreasing probability of activation after an initial activation. pM is given as:

$$pM = P(X > x) = 1 - \Phi\left(\frac{\ln(x) - \mu}{\sigma}\right) \quad (3)$$

where:

- $P(X > x)$ denotes the complementary cumulative distribution function.
- X denotes the log-normally distributed random variable.
- x denotes the specific value for which we are calculating the probability.
- μ denotes the mean of the underlying normal distribution of the natural logarithm of X .
- σ denotes the standard deviation of the underlying normal distribution of the natural logarithm of X .
- Φ denotes the cumulative distribution function (CDF) of the standard normal distribution.
- $\ln(x)$ denotes the natural logarithm of x .

Before an initial activation, the inter-event time since the agent’s last activation was not defined. pM was therefore, by default, set to 0 if an agent had not yet been activated during the simulation. This meant that the agents had to rely on pB for their first activation. After this initial activation, pB was bolstered by pM , effectively increasing their probability of another activation.

Since pM was modeled using the CCDF, its values decreased as the time since an agent’s last activation increased. In-depth instructions on how the log-normal distribution was fit to the data are given in Appendix C.

The final component of the activation function, the threshold T , is given as:

$$T \sim \mathcal{U}(0, 1) \quad (4)$$

where:

- T denotes a random variable.
- $\mathcal{U}$ denotes a uniform distribution.

It’s intentionally designed to be dynamic in order to allow all agents a chance of being activated, which is given by their base activation probability. Additionally, since pM is agent-specific and only takes values greater than zero after an agent has been activated for the first time, setting T to a constant value anywhere above 0.1 would have locked all agents from the simulation.

4.4 Analysis

The following documents the analysis methods employed to judge the simulation framework’s validity with regard to persona imprinting, social media interaction, and discussion networks.

4.4.1 Persona Imprinting

Overall activity measures were computed, showing the number of active agents who created at least one comment and those that did not comment and stayed silent. Along with this, the total volume of comments created was determined. These findings, split by individual threads – i.e., seed post discussions – were compared to the baseline data, allowing insights into which topics received which amount of attention by users and agents, respectively. Analogous to those actions available to agents within the simulation, counts of how often these actions had been executed by participants in the baseline experiment were approximated in case they were not contained in the baseline data. The number of comment likes and dislikes, for example, could not be extracted directly since the data only contained comment scores, i.e., the number of likes minus the number of dislikes. These values were approximated by summing up the scores of all comments with positive and negative

scores, respectively. The data was then compared to the action execution counts of the simulated data. Analyzing activity on the user/agent level, the mean and median volume of comments per user/agent were identified, and the correlation between an agent’s and its respective seed user’s comment volume was calculated. Toxicity scores of the individual comments were retrieved. In the baseline experiment, the Google Perspective API was used to retrieve toxicity scores. Following this, the Perspective API was queried for the toxicity scores of the simulated comments. The mean toxicity per simulated thread was compared to the mean toxicity in the baseline data, and a correlation analysis of mean toxicity per agent and their respective seed user was calculated. Finally, following the baseline experiment, I applied a combination of a zero-hurdle and a count model to the simulated data. The same predictors as in the baseline experiment were analyzed, with the exception of those extracted from the weekly check-up surveys in the experiment, since this data was not available for the simulation. The zero hurdle model was used to identify the factors that influenced an agent to become active – i.e., publish their first comment – or deterred them from doing so. The count model, on the other hand, was used to model the influence of different factors on the volume of comments published by an agent. The results of the zero hurdle and count model were then compared to the baseline data.

4.4.2 Social Media Interaction

Differences in inter-event time between seed users and agents were analyzed by comparing the respective inter-event time distributions. Additionally, per user/agent inter-event time autocorrelation analyses were conducted, and the simulation data was compared to the baseline data. Similar to the analysis of comment volume, the volume of replies over all threads and for each thread individually was investigated both for the simulation and baseline data. On top of that, correlation analyses between overall (comment) volume and volume of replies per thread were conducted, and simulation-based findings were compared to baseline data-based ones. Lastly, comment length – i.e., the number of characters in a comment – was analyzed. The Comment length distribution was determined for the simulation

and compared to the distribution of the baseline data. Correlation analyses between comment length and the length of a replying comment were conducted, and simulation results were compared to the baseline.

4.4.3 Discussion Networks

For the structural analysis of the discussion networks, each thread was handled individually. The network size, width, and depth of each thread were determined. So was network transitivity, i.e., the degree of interconnectedness between the different users/agents. The mean and median values over all threads were calculated, and the difference between baseline and simulation means was compared using two-sample t-tests. Additionally, distributions of network size, width, and depth were compared visually. And finally, the number of self-replies – i.e., loops in the discussion networks – per thread was extracted for both the simulated and baseline data. The mean and median number of self-replies over all threads were then compared.

5 Results

This section reports the results of my analyses on each of the three simulation aspects under investigation: persona imprinting, social media interaction, and discussion networks.

5.1 Persona Imprinting

In the baseline experiment, 63% of users contributed at least one comment. In contrast, the simulation showed a much higher participation rate, with 96% of agents commenting at least once. While 36% of users remained silent, only 4% of the agents did not comment in the simulation. The average share of active users per thread was 25%, compared to 38% for simulated agents. The median share of active users per thread was 24%, while the median for agents was 42%. Table 1 gives an overview including absolute participation counts. Two-sample t-test results between the two individual Subreddits showed no significant difference in means for per-topic share of active users/agents. This held for both the baseline ($p = 0.483$) as well as simulated data ($p = 0.772$). For detailed per-thread info, see Table D2.

The total volume of comments in the baseline data was 1,897, with 982 comments in the first

Table 1: Shares and counts of active/silent users/agents. Values for all threads combined and averaged per-thread are given. Simulated absolute values are averaged over all 10 runs and rounded to zero digits.

Participation	Baseline Data		Simulated Data	
	Share	Absolute	Share	Absolute
Active users/agents	0.63	114	0.96	173
Silent users/agents	0.37	66	0.04	7
Mean Active users/agents per thread	0.25	45	0.38	68
Median Active users/agents per thread	0.24	43	0.42	76

Subreddit and 915 in the second. The simulation runs generated a total of 3,899 ($SD = 50$) comments on average, distributed as 1,853 ($SD = 47$) in the first Subreddit and 2,046 ($SD = 63$) in the second. Correlation analyses revealed no significant relationship between the per-thread comment volumes of the two Subreddits in the baseline data ($p = 0.566$). However, a strong positive correlation ($r = 0.813$, $p < 0.001$) was observed in the simulated data. No significant correlation was found between baseline and simulated first ($p = 0.519$) and second ($p = 0.394$) Subreddit with regard to comment volume per thread. For detailed per-thread info, see Table D2.

The analysis of minimum and maximum comment scores showed a range from -4 to 22 in the baseline data, and from -1 ($SD = 0$) to 20 ($SD = 5$) on average in the simulation runs. The maximum number of comments by a single user was 87 in the baseline data, with a mean of 11 ($SD = 18$) and a median of 3 comments per user. In the simulation runs, the average maximum was 66 ($SD = 5$), with a mean of 22 ($SD = 14$) and a median of 20 comments per agent. Figure 2 shows the distribution of comment count for both the baseline and simulated data. On the individual level, there was a weak ($r = 0.153$ ($SD = 0.026$)), in parts significant ($p = 0.053$ ($SD = 0.043$)) effect of seed user comment count on agent comment count averaged over all simulation runs.

The count analysis of the different executed actions shows that users liked content significantly more often than agents. Users, on the other hand, created significantly fewer comments than simulated agents. Figure 3 shows the simulated average, min, and max values along with the approximated values for the baseline data. There are no values for the action `continue.browsing`

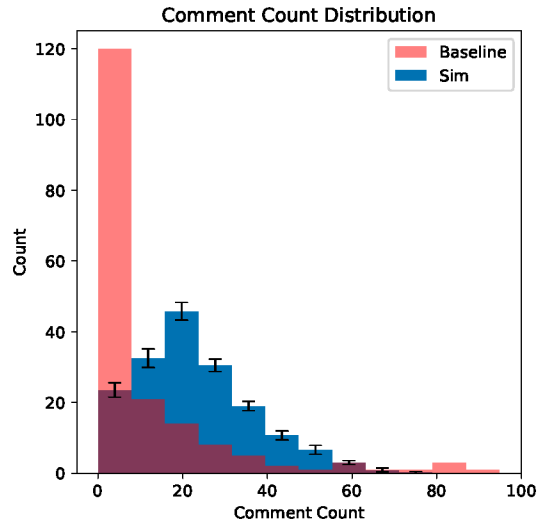


Fig. 2: Distribution of baseline vs. simulated comment count. Average values are shown for the simulated data with brackets indicating min and max values over all 10 simulation runs.

from the baseline data, since this was impossible to approximate given the available data. The values for `create.comment` were estimated by extracting the number of first-level comments, and those for `create.comment.comment` by determining the number of all second- or higher-level comments. Both like and dislike counts are cautious estimates, with their real values probably being much higher. Due to data availability, these approximations were the best I could provide.

The toxicity analysis revealed the simulated threads to be slightly more toxic than those from the baseline experiment. The averaged Perspective API toxicity score for the baseline threads was 0.103 ($SD = 0.043$), and the median was 0.048.

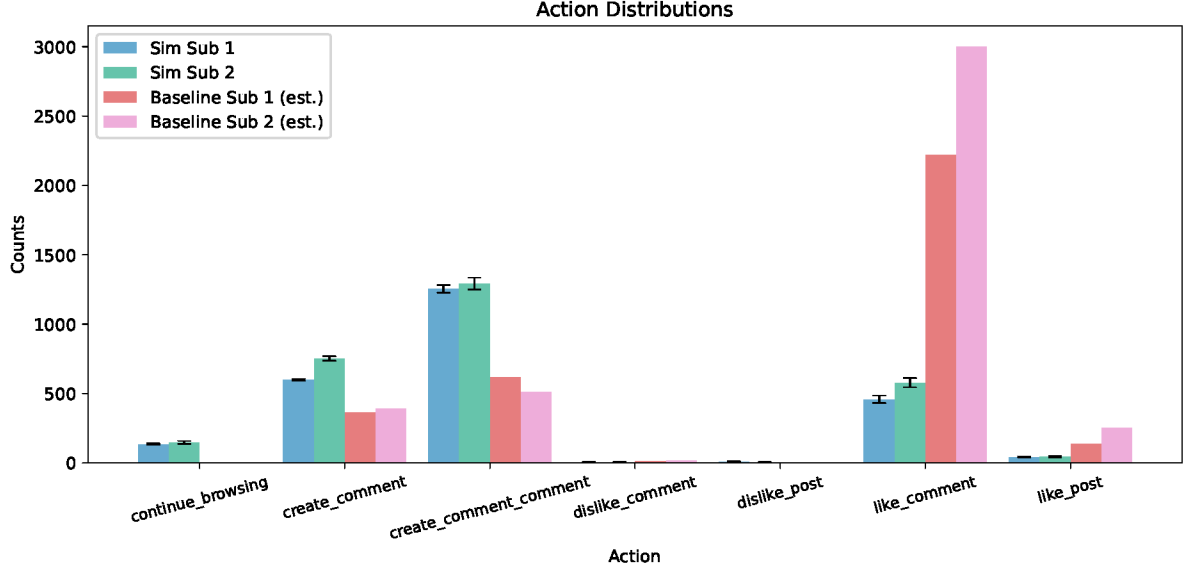


Fig. 3: Distributions of user/agent actions across the two Subreddits. Values for the baseline experiment are estimations where no data was available. No estimates are given for `continue_browsing`. For the simulated data values are averaged over 10 simulation runs with min and max values indicated by the brackets.

Over all simulation runs, the average value was 0.156 ($SD = 0.048$) and the median was 0.120. A detailed overview of per-topic toxicity is given in Table D2. A strong ($r = 0.718$ ($SD = 0.090$)), significant ($p = 0.003$ ($SD = 0.005$)) correlation was found between the baseline and simulated toxicity scores per thread. Seed users' comment toxicity, though, had no significant ($p = 0.360$ ($SD = 0.283$)) influence ($r = 0.093$ ($SD = 0.076$)), on their respective agent's comment toxicity averaged across the 10 simulation runs.

The zero-hurdle and count model analysis showed an over the simulation runs averaged significant effect of a seed user's self-reported commenting behavior on the probability of an agent getting active in the discussion. For already active agents, the same was true, and only the effect of a seed user's self-reported commenting behavior on the comment count of the agent remained robust against different model specifications. This is shown in Figure 4. For alternative model specifications, see SI.

5.2 Social Media Interaction

The inter-event time distributions for the baseline and simulated data are visualized in Figure 5. A bimodal shape was observed for both distributions. One peak at the low end of the distribution, and the other at about 80,000 seconds or 22 hours. Visual analysis suggests that the two distributions are fairly similar. A two-sample Kolmogorov-Smirnov test was significant ($p < 0.001$), thereby rejecting the null hypothesis that the two distributions were drawn from the same distribution. Individual level examples of IET between consecutive comments are shown in Figure D6 and Figure D7.

The autocorrelation analyses of users' inter-event time revealed a small negative averaged effect ($\rho_1 = -0.072$ ($SD = 0.255$)) of a comment's inter-event time on the inter-event time of the subsequent comment by the same user. The averaged results of Ljung-Box tests rejected the statistical significance of the autocorrelations ($p = 0.492$ ($SD = 0.301$)). For the simulated data, the averaged effect was negative ($\rho = -0.184$ ($SD = 0.175$)) and not significant ($p = 0.406$ ($SD =$

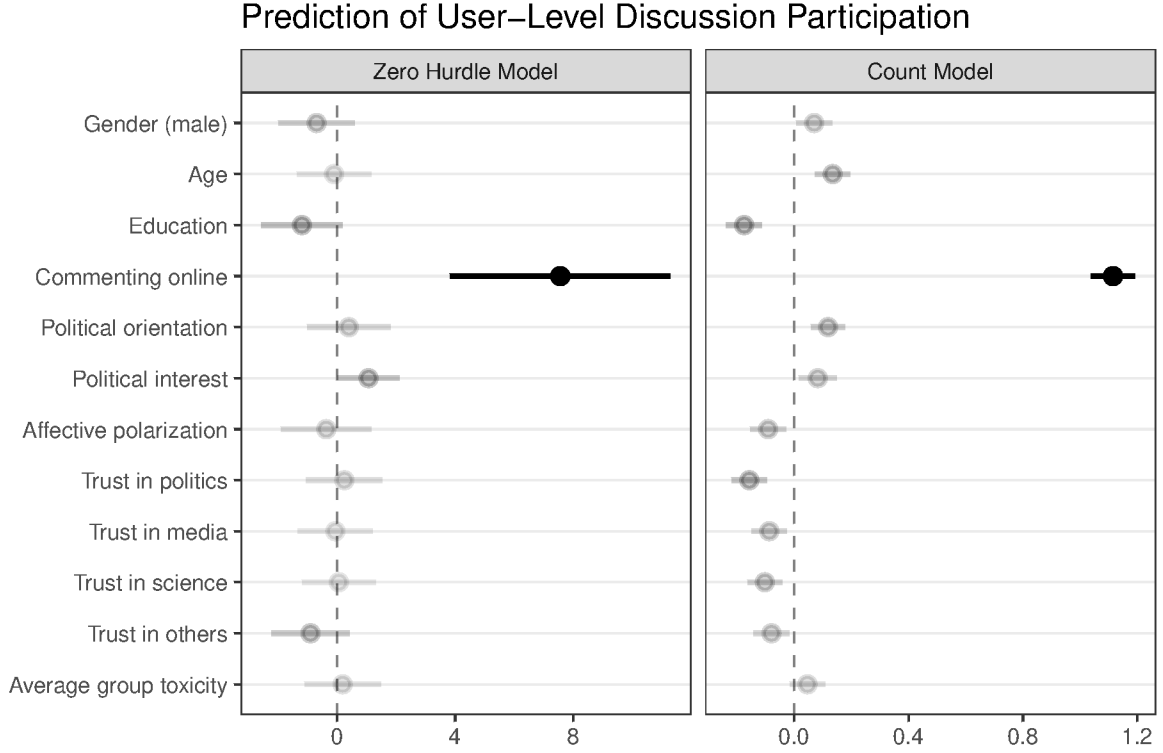


Fig. 4: Zero hurdle model of factors influencing agents to become active. Poisson count model of factors influencing agents’ comment count. For alternative model specifications and model robustness, see SI.

0.279)). Figure 6 visualizes current comment inter-event time vs. previous comment inter-event time for the baseline and simulated data.

The analysis of reply counts revealed that, on average, 28 ($SD = 25$) replies were created per thread in the baseline experiment. The median count was 24. For the simulation runs, the average lay at 12 ($SD = 4$) and the median at 11. A detailed per-topic overview is provided in Table D2. On the Subreddit-level, an average of 16 ($SD = 26$) and a median of 6 replies were created per thread in the first baseline Subreddit. In the second Sub the average reply count per thread was 12 ($SD = 10$) and the median was 12. The difference of means between the two baseline Subs was not significant ($p = 0.474$). For the first simulated Subreddit the average reply count per topic was 6 ($SD = 3$) and the median was 6. For the second Sub the average and median were 6 ($SD = 3$) and 6, respectively. The difference of means between the two simulated Subreddits was not significantly

different from 0 ($p = 0.323$ ($SD = 0.237$)) when averaged over all simulation runs.

A strong ($r = 0.838$), significant ($p < 0.001$) positive effect of comment volume per thread on reply volume per thread was found for the baseline data. For the simulation, the effect was negative ($r = -0.457$, $p = 0.043$). Reply volume per thread of the baseline and simulated data was not significantly correlated ($p = 0.162$).

For both the baseline and simulated data, the distributions of comment length were log-normally shaped as shown in Figure 7. In both cases, the correlation between a comment’s length and the length of a reply was statistically significant ($p < 0.001$). For the simulation runs, though, the averaged effect was less pronounced ($r = 0.195$ ($SD = 0.134$)) than the effect in the baseline data ($r = 0.331$).

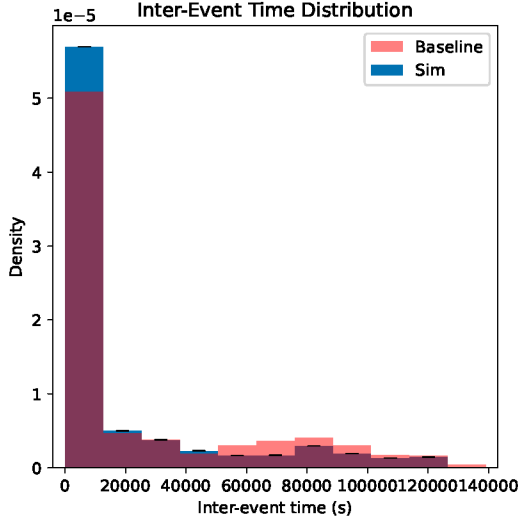


Fig. 5: Distribution of baseline vs. simulated IET. Average values are shown for the simulated data with brackets indicating min and max values over all 10 simulation runs.

5.3 Discussion Networks

Figure 8 shows the distributions of discussion network size, width, and depth of both baseline and simulated data. Two-sample t-tests of difference of means between baseline and simulated data were significant for size ($p < 0.001$ ($SD < 0.001$)), width ($p < 0.001$ ($SD < 0.001$)), and depth ($p = 0.002$ ($SD = 0.001$)) when averaged over all simulation runs. In the baseline data network size and width ($r = 0.641$, $p < 0.001$), size and depth ($r = 0.767$, $p < 0.001$), and width and depth ($r = 0.351$, $p = 0.026$), were significantly positively correlated. In the simulation runs, size and width were positively ($r = 0.955$ ($SD = 0.012$)) correlated ($p < 0.001$ ($SD < 0.001$)). Size and depth ($r = -0.249$ ($SD = 0.170$), $p = 0.214$ ($SD = 0.216$)) and width and depth ($r = -0.356$ ($SD = 0.169$), $p = 0.141$ ($SD = 0.29$)) were not significantly correlated.

Mean network transitivity over all threads from the baseline experiment was 0.098 ($SD = 0.065$) and median 0.087. For the simulated data, the mean and median lay at 0.255 ($SD = 0.139$) and 0.210, respectively. The difference of means was significant ($p = 0.024$ ($SD = 0.010$)).

Concerning network loops or self-replies per-thread, the baseline data mean and median were 0 ($SD = 0$) and 0. The simulation data mean was 4 ($SD = 0$) and the median 4. The difference of means between baseline and simulated self-replies per thread was significant ($p < 0.001$ ($SD < 0.001$)) when averaged across the 10 simulation runs. For a visualization of each of the baseline experiment’s thread structures as well as those of the simulated threads, see SI.

6 Discussion

6.1 Persona Imprinting

Agents imprinted with the persona data of human seed users over-portray commenting behavior while downplaying other behaviors. This seems to be caused by the agents’ reluctance to keep silent when also offered the opportunity to voice their opinions. The over-eager tendency of LLMs to act has already been uncovered in LLM-based simulations of other domains [1, 29]. The results of my analysis support this by showing that the simulated agents were much more active than human users. Not only was the share of active participants much higher in the simulation, but also the total volume of created comments. The analysis on the thread level led to the same results, with simulated threads receiving much more engagement than baseline ones. Per individual analysis showed the median comment count per agent to be more than six times higher than in the baseline data, and the weak and mostly non-significant effect of seed user comment count on agent comment count further supports this claim. The action distribution also revealed that agents chose to create a comment about 10 times as often as they chose to continue browsing. Compared to the baseline data, they chose to like content about five times less. So, agents were clearly biased toward actions that created comments instead of recreating the more passive behaviors, i.e., by continuing to browse or liking/disliking content.

The effect of the simulation framework’s activation function seems to successfully model agents being on- and offline. While agent activity was high, this seems to have been caused by a single factor: their seed user’s self-reported online commenting behavior. Research showed that among active users, self-reported online commenting

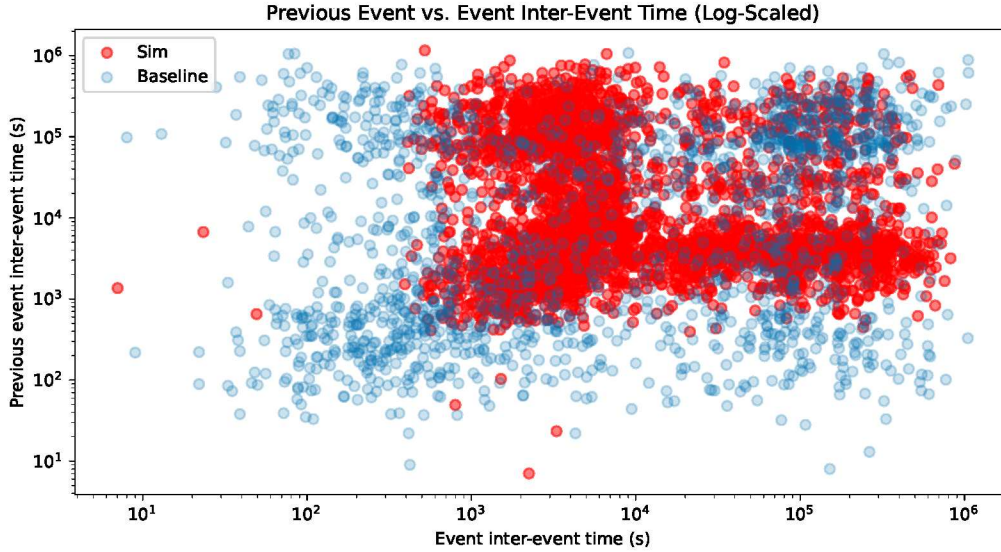


Fig. 6: Scatter plot of a comment’s IET vs. the IET of the previous comment by the same user/agent. Both axes are log-transformed. The plot visualizes the four groups: short IET followed by short IET, short IET followed by long IET, long IET followed by short IET, long IET followed by long IET.

behavior significantly influenced the number of comments created [34]. In line with this, the count model analysis of the simulation showed that a seed user’s self-reported online commenting behavior positively influenced the number of comments created by the respective agent. Contrary to this, the probability of agents becoming active was also influenced by their seed users’ self-reported online behavior. In the baseline experiment, self-reported commenting behavior did not significantly influence a user’s choice to become active. It seems, therefore, likely that the activation function correctly models an individual’s online/offline state, providing the agents with windows of opportunity to act. The over-eagerness of the agents to produce comments, though, causes them to fill these windows by creating comments. This, in turn, results in the significant effect of seed user self-reported commenting behavior on participation and comment count. Other significant factors identified in the baseline experiment did not significantly influence activity in the simulation, leading me to further conclude that the agents do not correctly translate their persona data into online behavior.

Simulation agents do not accurately infer topic relevance from the given persona data. This is

shown by the per-Subreddit analysis. While per-topic engagement between the two Subs was uncorrelated in the baseline data, the simulated Subreddits showed a strong correlation with each other. Additionally, no significant effect of per-thread engagement between the baseline and simulated Subreddits was detected. This suggests that the agents did not base their choices to engage with specific seed posts on their persona data. For that to be the case, I would have expected the individual Subreddits, populated with different users/agents, to be correlated across baseline and simulated data, and the simulation to mimic the between-Subreddit difference apparent in the baseline data. I, therefore, assume that the agents chose to engage with topics based on an LLM-internal bias towards some of them. Another explanation could be that the agents did not simply mimic their seed user’s behavior but transformed it in ways that reflect structural bias [32, 49], with this bias then governing their decisions to engage. It should be noted, however, that per-thread engagement in the simulations is highly dependent on the pre-simulation layer, where agents are shown the seed post and asked to either participate in the discussion or continue

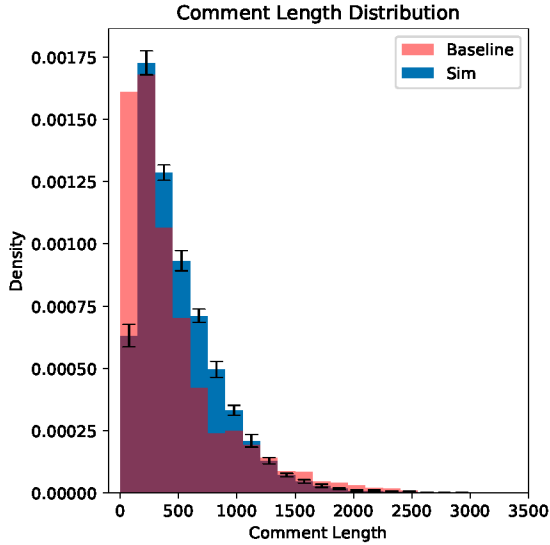


Fig. 7: Distribution of baseline vs. simulated comment length. Average values are shown for the simulated data with brackets indicating min and max values over all 10 simulation runs.

browsing. In a simulation design where the different threads were simulated side-by-side, the engagement dynamics would certainly differ.

Agents do not reliably translate persona data into toxic behavior but accurately reproduce toxic trends for specific topics. The results of my simulation’s toxicity analysis support this claim. The very weak and non-significant effect of a seed user’s average toxicity over all comments on their respective agent’s toxicity suggests that the persona data did not influence toxic agent behavior. It rather seems that certain topics offer more toxic potential than others, and that both users and agents therefore tend to show more toxic behavior when discussing them. The strong correlation of per-topic toxicity between the baseline and simulated discussions supports this. Thereafter, topic discussions that were, on average, more toxic between users were more toxic when simulated, too. This notion of topic-dependence when it comes to toxicity has also been identified by others. Especially, political conversations tend to elicit toxic behavior [41, 42], and some of the seed posts clearly are more political than others. An example of this would be: “The US should condemn Israel’s military actions in Gaza as acts of genocide and impose full sanctions.” vs.

“Airbnb should be banned in cities.”. While the agents seem to capture these topic-specific toxicity trends, they also tend to exaggerate toxic behavior. The overall toxicity in the simulations was higher than what was observed in the baseline data. Interpreting the results according to the Perspective API documentation², 10 out of 100 readers would classify a random comment from the baseline data as toxic. For the simulated comments, 16 out of 100 readers would do so. I suspect the increased toxicity might have been caused by the prompts specifically instructing the agents to show anti-social behavior. It might also have been caused by the persona imprinting, as research suggests that providing LLMs with more context info can lead to increased harmful language use [32]. Running the simulation without persona imprinting and/or the mentioned instructions could lead to further insights.

6.2 Social Media Interaction

The simulation framework closely reproduces comment inter-event time trends and windows of burst-like agent action, followed by spans of inactivity, which were found in human online commenting behavior [30, 51, 56]. This is supported by the simulated distribution of IET, which closely follows that of the baseline data. The bimodal nature of the distribution supports the assumption that two separate processes are at play: bursts and between-burst downtimes. Research on human online communication interaction has identified this pattern for email [7, 17, 52], short-message [50], and social media communication [10, 27, 56]. While the baseline and simulated IET distributions were not drawn from the exact same distribution, as evidenced by the results of a Kolmogorov-Smirnov test, this does not discredit the successful reproduction of burstiness. The general patterns were mimicked but not equally distributed 1:1. The autocorrelation analysis furthermore supports the simulation’s ability to reproduce human online interaction patterns. The non-significant and close to zero effect sizes for baseline and simulated data imply that commenting is a memory-less process. IET is not affected by the IET of the previous event. In combination with the burstiness indicated by the

²Perspective API Documentation

Discussion Network Size, Width, and Depth Distribution

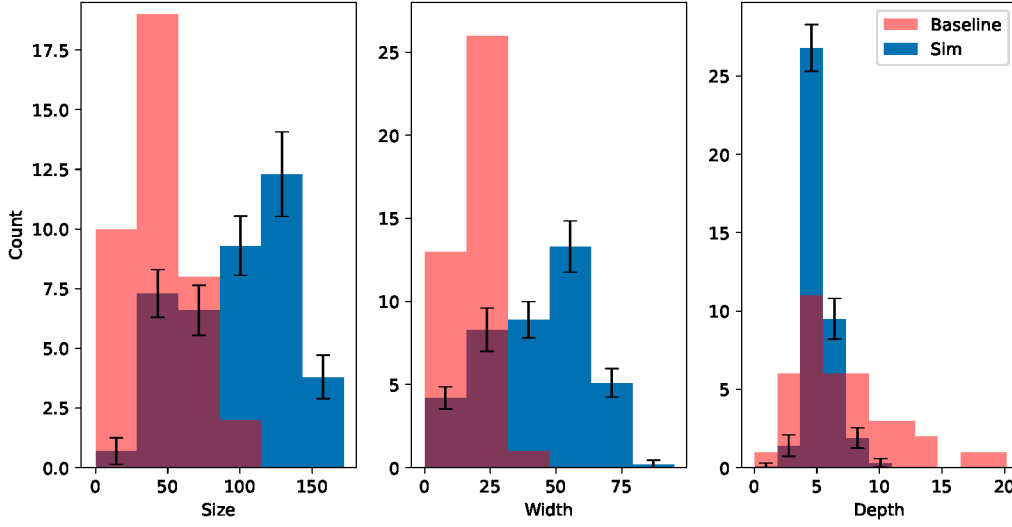


Fig. 8: Distributions of baseline vs. simulated discussion network size, depth, and width. Average values are shown for the simulated data with brackets indicating min and max values over all 10 simulation runs.

IET distributions, this suggests an on-off state of interaction where once activated, comments are created seemingly at random [17]. For the individuals, this means that their commenting or not is dependent on their state – i.e., within-burst vs. between-burst – instead of on the IET of their last interaction.

Along those lines, Figure 6 shows partially contradicting results. The IET in the baseline experiment can be roughly divided into four groups. First, short IET preceded by short IET: these are comments that were created within a burst. Then, short IET preceded by long IET: comments which mark the beginning of a burst at least two events long. Third, long IET preceded by short IET: these mark comments created after a burst, with long downtime in between. Lastly, long IET preceded by long IET: these are sporadic comments, most likely by users who just comment every few days. Figure 6 indicates that the majority of user comments were created either in bursts spanning about 2.5 hours and spaced 12-24 hours apart or sporadically every roughly 12 to 72 hours. This indicates that user interaction was probably governed by daily schedules. The IET grouping wasn't recreated as clearly by the simulation. Most strikingly, agents created fewer comments sporadically. This is especially interesting since the agents

created significantly more comments. Given their previously identified over-eagerness to comment, it's not surprising, though. While users might view Reddit multiple times a day but remain silent most of the time, agents that were simulated to be online this often, nearly always commented. With regard to the other three IET groupings, the simulation showed similar trends as the baseline, but the divides between the three were less distinct. This might in part be explained by agents' lack of daily routines and agendas. Users, for example, may have viewed the platform in the morning before work and then again in the early evening, as a result of their daily schedules. The activation function, on the other hand, did not take these periodic effects into account. While agents were modeled to reproduce burstiness, the start of these bursts was modeled to randomly occur over the course of the simulation. This explains the lack of clear divides between the three non-sporadic IET clusters observed in the baseline data.

Agents under-represent the typical back-and-forth between Reddit users and struggle to keep their sense of self with growing platform environment. This is shown by the analysis of reply counts. There were more than twice as many replies in the baseline data, even though the total

volume of comments was only half that of the simulated data. This suggests a clear lack of agents engaging in typical back-and-forths or arguments with others, as shown by users. For visualizations of the discussion networks documenting this behavior, see SI. The correlation analysis between per-thread comment volume and reply volume showed a strong positive and significant effect for the baseline data. This was expected as it shows that the number of replies created grew with the overall number of comments created. Users did not just create unrelated comments but engaged in sub-discussions within the individual threads. For the simulated data, the effect of comment volume on reply count per topic was negative and significant. This supports the claim that agents had trouble retaining a sense of self when the number of comments per thread increased. Instead of continuing to expand on the present back-and-forths, agents lost track and commented on other content.

These effects of under-representation and self-loss are possibly confounded by the design of the simulation, though. Reddit usually sorts the content within a thread according to an algorithm. This generally increases the likelihood of the most popular comments being seen by users, making them even more popular. This is also known as ranking bias effect and has been well documented by research across different domains [15, 16, 22, 25, 31, 40]. On Reddit specifically, research found that the probability of users interacting with content depended on content positioning and score [15], which is directly tied to positioning through the recommendation algorithm. The simulation employed no recommendation algorithm, and in each simulation step, agents were provided with the entire thread. It is therefore likely that this difference in platform presentation has influenced the agents’ reply behavior, for example, by causing them to engage with all content rather than mostly engaging in specific arguments with other agents. In addition to that, the mechanics behind notifications might have influenced user behavior. The Reddit mobile app, at least, informs users in real-time about replies to their comments and navigates them directly to the reply. Therefore, there is no need to engage with other content when replying. Distractions are removed, so to speak. The agents, on the other hand, viewed the entire thread. It is possible that this distraction caused them to engage with other content instead

of continuing a previous back-and-forth. Finally, it also seems likely that not the simulation design necessarily, but rather the LLM’s memory, so to speak, caused the downplay of reply behavior, and that rather than losing track of themselves, the agents lost track of the previous discussion. Research has shown that positioning of key information, especially within long sequences, dramatically influences an LLM’s capacity to faithfully recall it [28]. It, therefore, grew increasingly harder for the agents to maintain critical information on who wrote which comment when the length of the threads increased. This might have caused them to stop basing their decisions on where to comment on authorship information. To shed more light on this, additional research is needed. An investigation of agent reply behavior during the first few simulation steps, where the environment would not be as complex yet, might be especially informative.

The simulation framework reproduces the trend of the baseline comment length distribution and replies to mimic their target’s length. This is limited by a discrepancy among short (< 200 characters or ~ 43 words) comments, whereof significantly more are found in the baseline data. The agents were specifically instructed on how long individual comments should be and to mimic the target comment’s length. Thus, this finding isn’t surprising per se. When put into context with other instructions that were not followed, though, it gains relevance. As mentioned, the agents did not seem to keep track of who they were – i.e., their ID – during the simulation, and they also did not follow the instruction to avoid self-replies, as I’ll discuss below. Considering that all these instructions were placed at the front of the prompt, this finding thus implies that agents likely did not lose track of who they were but rather of what they had already done. They memorized the instructions placed at the beginning of the prompt. Information on comment authorship, which was given considerably later in the prompt, was forgotten or falsely remembered. This is in line with previous research [28]. The simulation would therefore likely profit from additional measures aimed at ensuring that key information is correctly retained by agents.

6.3 Discussion Networks

The simulation framework overestimates discussion network size and width while underestimating depth. This is indicated by the findings presented in Figure 8. Possible explanations for this have already been discussed, albeit in different contexts. In this case, the observed results were caused by the interplay of agents’ over-eagerness to comment and their lack of participation in between-agent back-and-forths. This is supported by the findings of the correlation analysis. While both width and depth positively correlated with size for the baseline data, only size and width positively correlated in the simulation. This indicates that instead of a holistically growing network – i.e., size, width, and depth grow together – as observed in the baseline data, the simulation produced a network that simply widened with increasing size. In the simulation, the correlation between size and depth was negative, though not significant, implying that agents lost track of important information needed to create depth as the thread grew. Baseline and simulation network transitivity means were significantly different from each other. This shows that agents were either less discerning in whom they interacted with, or it solidifies the assumption that agents were unable to keep track of important environment information and therefore chose where to comment according to an unknown factor. Given the already discussed findings, the second seems much more likely, though only further research can bridge this knowledge gap.

Finally, my research showed that self-replies or network loops per thread appeared, on average, about 26 times as often in the simulated compared to the baseline threads. The difference in means between simulation and baseline data was significant. The most likely explanation for this is tied to previous findings. Due to the increasing complexity of the environment, agents lost track of or misremembered the authorship information of previous comments. This led them to reply to themselves, most likely not realizing that they had authored the comment they were replying to. An easy fix to prevent such behavior would be to programmatically prohibit self-replies. This would not be in line with Reddit’s platform design, though, and would not address the underlying

mechanism that caused agents to so frequently reply to themselves.

7 Limitations

As previously mentioned, the proposed simulation framework does not simulate the contest between different threads, let alone Subreddits. Combined with the over-eagerness of the agents to comment, this certainly led to overly high engagement in all threads. As of now, the pre-simulation layer only prompts agents to either join a thread or continue browsing for other content. This other content is not defined, though. Doing so, by providing the model with a rich selection of possible threads to engage with, could help to broaden the framework’s external validity.

Another limiting factor, not mentioned yet, concerns OASIS. As of version 0.1.1 – the basis for the present framework – agents are prompted simultaneously at the start of an environment step. Since the prompt contains the thread state, all agents are presented with the same content. While the first agents have already executed their action, other agents are still working to do so. The additional information of the already executed actions is not conveyed to the agents anymore, since the prompts have already been sent. Agents are thus working with outdated information. This might very well have confounded the results regarding interaction and network structure. A possible solution could involve modeling agent priority and then simulating the activated agents one after another.

Finally, one must always be mindful of the possibility that individuals or groups may be misrepresented when employing LLMs to substitute humans [39, 43, 49]. LLM outputs may be highly sensitive to the wording of prompts [9], and these models’ black-box nature limits the degree to which research can retrace and understand emergent behavior [23].

8 Conclusion

This work presents a customized version of the OASIS framework, specially created for the simulation of individual Reddit threads using LLM-based agents. Utilizing this framework, I conducted, to my knowledge, the first simulation of Reddit threads where the agents were responsible

not only for the generation of content but also for action decision-making. The analysis of the simulation showed clear differences between the baseline and simulated behavior. Agents showed a general over-eagerness to create content and were less discerning about whom to reply to compared to human users. This led to overly large and wide discussion networks lacking depth. At the same time, the simulation framework closely reproduced the burst-like patterns of interaction and distribution of comment length observed with human users. While failing to simulate individual-level toxic behavior, the simulation slightly overestimated per-topic toxicity but clearly recreated baseline trends. Generally speaking, the statistics- and research-based aspects of the simulation showed clear potential while those governed by the LLM were of mixed success. Future research into this topic might therefore profit from a more conventional ABM-like approach of simulation, not just modeling agent activation but also action, and further restricting the LLM’s influence on the outcome.

Supplementary information.

- The simulation framework is available at <https://github.com/rutschmanna/oasis>
- Supplementary materials and Data are available at <https://github.com/rutschmanna/gtfoou>

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This is for me, but it is because of you. Love you!

References

- [1] Aher G, Arriaga RI, Kalai AT (2023) Using large language models to simulate multiple humans and replicate human subject studies. <https://doi.org/10.48550/arXiv.2208.10264>, arXiv:2208.10264
- [2] Alstott J, Bullmore E, Plenz D (2014) powerlaw: A python package for analysis of heavy-tailed distributions. PLOS ONE 9(1):1–11. <https://doi.org/10.1371/journal.pone.0085777>
- [3] Aragón P, Gómez V, García D, et al (2017) Generative models of online discussion threads: state of the art and research challenges. Journal of Internet Services and Applications 8(1):15. <https://doi.org/10.1186/s13174-017-0066-z>
- [4] Argyle LP, Busby EC, Fulda N, et al (2023) Out of one, many: Using language models to simulate human samples. Political Analysis 31(3):337–351. <https://doi.org/10.1017/pan.2023.2>
- [5] Arthur C (2006) What is the 1% rule? URL <https://www.theguardian.com/technology/2006/jul/20/guardianweeklytechnologysection2>
- [6] Bail C (2021) Breaking the Social Media Prism: How to Make Our Platforms Less Polarizing. Princeton University Press, <https://doi.org/10.1515/9780691216508>
- [7] Barabási AL (2005) The origin of bursts and heavy tails in human dynamics. Nature 435(7039):207–211. <https://doi.org/10.1038/nature03459>
- [8] Barberá P (2015) Birds of the same feather tweet together: Bayesian ideal point estimation using Twitter data. Political Analysis 23(1):76–91. <https://doi.org/10.1093/pan/mpu011>
- [9] Bisbee J, Clinton JD, Dorff C, et al (2024) Synthetic replacements for human survey data? the perils of large language models. Political Analysis 32(4):401–416. <https://doi.org/10.1017/pan.2024.5>
- [10] Blenn N, Mieghem PV (2016) Are human interactivity times lognormal?

- <https://doi.org/10.48550/arXiv.1607.02952>, [arXiv:1607.02952](https://arxiv.org/abs/1607.02952)
- [11] Brown TB, Mann B, Ryder N, et al (2020) Language models are few-shot learners. In: Proceedings of the 34th International Conference on Neural Information Processing Systems. Curran Associates Inc., Red Hook, NY, USA, NIPS '20, URL <https://dl.acm.org/doi/abs/10.5555/3495724.3495883>
 - [12] Cinelli M, Morales GDF, Galeazzi A, et al (2021) The echo chamber effect on social media. Proceedings of the National Academy of Sciences 118(9):e2023301118. <https://doi.org/10.1073/pnas.2023301118>
 - [13] Ferraro A, Galli A, La Gatta V, et al (2025) Agent-based modelling meets generative ai in social network simulations. In: Aiello LM, Chakraborty T, Gaito S (eds) Social Networks Analysis and Mining. Springer Nature Switzerland, Cham, pp 155–170, https://doi.org/10.1007/978-3-031-78541-2_10
 - [14] Gao C, Lan X, Lu Z, et al (2023) S3: Social-network simulation system with large language model-empowered agents. <https://doi.org/10.48550/arXiv.2307.14984>, [arXiv:2307.14984](https://arxiv.org/abs/2307.14984)
 - [15] Glenski M, Weninger T (2017) Predicting user-interactions on reddit. In: Proceedings of the 2017 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining 2017. Association for Computing Machinery, New York, NY, USA, ASONAM '17, pp 609–612, <https://doi.org/10.1145/3110025.3120993>
 - [16] Glenski M, Weninger T (2017) Rating effects on social news posts and comments. ACM Trans Intell Syst Technol 8(6). <https://doi.org/10.1145/2963104>
 - [17] Goh KI, Barabási AL (2008) Burstiness and memory in complex systems. EPL (Europhysics Letters) 81(4):48002. <https://doi.org/10.1209/0295-5075/81/48002>
 - [18] Gonzalez-Bailon S, Kaltenbrunner A, Banchs RE (2010) The structure of political discussion networks: A model for the analysis of online deliberation. Journal of Information Technology 25(2):230–243. <https://doi.org/10.1057/jit.2010.2>
 - [19] Gu C, Luo L, Zaidi ZR, et al (2025) Large language model driven agents for simulating echo chamber formation. <https://doi.org/10.48550/arXiv.2502.18138>, [arXiv:2502.18138](https://arxiv.org/abs/2502.18138)
 - [20] Gómez V, Kappen HJ, Litvak N, et al (2013) A likelihood-based framework for the analysis of discussion threads. World Wide Web 16(5):645–675. <https://doi.org/10.1007/s11280-012-0162-8>
 - [21] Horton JJ (2023) Large language models as simulated economic agents: What can we learn from homo silicus? <https://doi.org/10.48550/arXiv.2301.07543>, [arXiv:2301.07543](https://arxiv.org/abs/2301.07543)
 - [22] Joachims T, Granka L, Pan B, et al (2005) Accurately interpreting clickthrough data as implicit feedback. In: Proceedings of the 28th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval. Association for Computing Machinery, New York, NY, USA, SIGIR '05, pp 154–161, <https://doi.org/10.1145/1076034.1076063>
 - [23] Larooij M, Törnberg P (2025) Do large language models solve the problems of agent-based modeling? a critical review of generative social simulations. <https://doi.org/10.48550/arXiv.2504.03274>, [arXiv:2504.03274](https://arxiv.org/abs/2504.03274)
 - [24] Lazer DMJ, Baum MA, Benkler Y, et al (2018) The science of fake news. Science 359(6380):1094–1096. <https://doi.org/10.1126/science.aao2998>
 - [25] Lerman K, Hogg T (2014) Leveraging position bias to improve peer recommendation. PLOS ONE 9(6):1–8. <https://doi.org/10.1371/journal.pone.0098914>
 - [26] Li G, Hammoud HAAK, Itani H, et al (2023) CAMEL: Communicative agents for "mind" exploration of large language model society.

- <https://doi.org/10.48550/arXiv.2303.17760>,
arXiv:2303.17760
- [27] Li Z, Tang X, Zhou H, et al (2018) An empirical investigation and theoretic modeling for the collective online visiting behaviors. *Physica A: Statistical Mechanics and its Applications* 503:969–980. <https://doi.org/10.1016/j.physa.2018.08.090>
 - [28] Liu NF, Lin K, Hewitt J, et al (2024) Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics* 12:157–173. https://doi.org/10.1162/tacl.a_00638
 - [29] Lu Y, Huang J, Han Y, et al (2025) Prompting is not all you need! evaluating llm agent simulation methodologies with real-world online customer behavior data. <https://doi.org/10.48550/arXiv.2503.20749>, arXiv:2503.20749
 - [30] Luo G, Zhang Z, Diao S (2022) Empirical analysis and modelling social network user interaction behavior and time characteristics based on selection preference. *Information Sciences* 608:1202–1220. <https://doi.org/10.1016/j.ins.2022.07.021>
 - [31] Muchnik L, Aral S, Taylor SJ (2013) Social influence bias: A randomized experiment. *Science* 341(6146):647–651. <https://doi.org/10.1126/science.1240466>
 - [32] Nudo J, Pandolfo ME, Loru E, et al (2025) Generative exaggeration in llm social agents: Consistency, bias, and toxicity. <https://doi.org/10.48550/arXiv.2507.00657>, arXiv:2507.00657
 - [33] Oswald L, Munzert S, Barbera P, et al (2022) Beyond the tip of the iceberg? exploring characteristics of the online public with digital trace data. <https://doi.org/10.31235/osf.io/yfmzh>
 - [34] Oswald L, Schulz W, Lorenz-Spreen P (2025) A collective field experiment disentangling participation in online political discussions. <https://doi.org/10.31235/osf.io/p2jaq.v1>
 - [35] Park JS, Popowski L, Cai C, et al (2022) Social simulacra: Creating populated prototypes for social computing systems. In: *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery, New York, NY, USA, UIST ’22, pp 1–18, <https://doi.org/10.1145/3526113.3545616>
 - [36] Park JS, Seering J, Bernstein MS (2022) Measuring the prevalence of anti-social behavior in online communities. *Proc ACM Hum-Comput Interact* 6(CSCW2):1–29. <https://doi.org/10.1145/3555552>
 - [37] Park JS, O’Brien J, Cai CJ, et al (2023) Generative agents: Interactive simulacra of human behavior. In: *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery, New York, NY, USA, UIST ’23, pp 1–22, <https://doi.org/10.1145/3586183.3606763>
 - [38] Park JS, Zou CQ, Shaw A, et al (2024) Generative agent simulations of 1,000 people. <https://doi.org/10.48550/arXiv.2411.10109>, arXiv:2411.10109
 - [39] Plaza-del-Arco FM, Cercas Curry A, Curry A, et al (2024) Angry men, sad women: Large language models reflect gendered stereotypes in emotion attribution. In: Ku LW, Martins A, Srikumar V (eds) *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, Bangkok, Thailand, pp 7682–7696, <https://doi.org/10.18653/v1/2024.acl-long.415>
 - [40] Salganik MJ, Dodds PS, Watts DJ (2006) Experimental study of inequality and unpredictability in an artificial cultural market. *Science* 311(5762):854–856. <https://doi.org/10.1126/science.1121066>
 - [41] Salminen J, Almerexhi H, Milenković M, et al (2018) Anatomy of online hate: Developing a taxonomy and machine learning models for identifying and classifying hate in online news media. *Proceedings of the International*

- AAAI Conference on Web and Social Media 12(1). <https://doi.org/10.1609/icwsm.v12i1.15028>
- [42] Salminen J, Sengün S, Corporan J, et al (2020) Topic-driven toxicity: Exploring the relationship between online toxicity and news topics. *PloS one* 15:e0228723. <https://doi.org/10.1371/journal.pone.0228723>
- [43] Santurkar S, Durmus E, Ladhak F, et al (2023) Whose opinions do language models reflect? In: *Proceedings of the 40th International Conference on Machine Learning*. JMLR.org, ICML’23, pp 29971–30004, <https://doi.org/10.5555/3618408.3619652>
- [44] Saveski M, Roy B, Roy D (2021) The structure of toxic conversations on Twitter. In: *Proceedings of the Web Conference 2021*. ACM, pp 1086–1097, <https://doi.org/10.1145/3442381.3449861>
- [45] Sobkowicz P, Thelwall M, Buckley K, et al (2013) Lognormal distributions of user post lengths in internet discussions - a consequence of the Weber-Fechner law? *EPJ Data Science* 2(1):2. <https://doi.org/10.1140/epjds14>
- [46] Sun N, Rau PPL, Ma L (2014) Understanding lurkers in online communities: A literature review. *Computers in Human Behavior* 38:110–117. <https://doi.org/10.1016/j.chb.2014.05.022>
- [47] Tumasjan A, Sprenger T, Sandner P, et al (2010) Predicting elections with Twitter: What 140 characters reveal about political sentiment. *Proceedings of the International AAAI Conference on Web and Social Media* 4(1):178–185. <https://doi.org/10.1609/icwsm.v4i1.14009>
- [48] Törnberg P, Valeeva D, Uitermark J, et al (2023) Simulating social media using large language models to evaluate alternative news feed algorithms. <https://doi.org/10.48550/arXiv.2310.05984>, [arXiv:2310.05984](https://arxiv.org/abs/2310.05984)
- [49] Wang A, Morgenstern J, Dickerson JP (2025) Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nature Machine Intelligence* 7:400–411. <https://doi.org/10.1038/s42256-025-00986-z>
- [50] Wei H, Xiao-Pu H, Tao Z, et al (2009) Heavy-tailed statistics in short-message communication. *Chinese Physics Letters* 26(2):028902. <https://doi.org/10.1088/0256-307X/26/2/028902>
- [51] Wu Y, Zhou C, Chen M, et al (2010) Human comment dynamics in on-line social systems. *Physica A: Statistical Mechanics and its Applications* 389(24):5832–5837. <https://doi.org/https://doi.org/10.1016/j.physa.2010.08.049>
- [52] Wu Y, Zhou C, Xiao J, et al (2010) Evidence for a bimodal distribution in human communication. *Proceedings of the National Academy of Sciences* 107(44):18803–18808. <https://doi.org/10.1073/pnas.1013140107>
- [53] Yang A, Li A, Yang B, et al (2025) Qwen3 technical report. <https://doi.org/10.48550/arXiv.2505.09388>, [arXiv:2505.09388](https://arxiv.org/abs/2505.09388)
- [54] Yang Z, Zhang Z, Zheng Z, et al (2024) OASIS: Open agent social interaction simulations with one million agents. <https://doi.org/10.48550/arXiv.2411.11581>, [arXiv:2411.11581](https://arxiv.org/abs/2411.11581)
- [55] Yu Y, Jiang J, Dhillon PS (2024) Characterizing the structure of online conversations across Reddit. *Proc ACM Hum-Comput Interact* 8(CSCW2):1–23. <https://doi.org/10.1145/3686913>
- [56] Zhang ST, Yuan HY, Duan LL (2020) Analysis of human behavior statistics law based on WeChat Moment. *Physica A: Statistical Mechanics and its Applications* 540:122854. <https://doi.org/https://doi.org/10.1016/j.physa.2019.122854>